

Exogenous vs. endogenous attention: Shifting the balance of fronto-parietal activity

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ABSTRACT

Despite behavioral and electrophysiological evidence for dissociations between endogenous (voluntary) and exogenous (reflexive) attention, fMRI results have yet to consistently and clearly differentiate neural activation patterns between these two types of attention. This study specifically aimed to determine whether activity in the dorsal fronto-parietal network differed between endogenous and exogenous conditions. Participants performed a visual discrimination task in endogenous and exogenous attention conditions while undergoing fMRI scanning. Analyses revealed robust and bilateral activation throughout the dorsal fronto-parietal network for each condition, in line with many previous results. In order to investigate possible differences in the balance of neural activity within this network with greater sensitivity, *a priori* regions of interest (ROIs) were selected for analysis, centered on the frontal eye fields (FEF) and intraparietal sulcus (IPS) regions identified in previous studies. The results revealed a significant interaction between region, condition, and hemisphere. Specifically, in the left hemisphere, frontal areas were more active than parietal areas, but only during endogenous attention. Activity in the right hemisphere, in contrast, remained relatively consistent for these regions across conditions. Analysis of this activity over time indicates that this left-hemispheric regional imbalance is present within the FEF early, at 3–6.5 s post-stimulus presentation, whereas a regional imbalance in the exogenous condition is not evident until 6.5–8 s post-stimulus presentation. Overall, our results provide new evidence that although the dorsal fronto-parietal network is indeed associated with both types of attentional orienting, regions of the network are differentially engaged over time and across hemispheres depending on the type of attention.

1. Introduction

Attention is an important selection mechanism that filters out irrelevant stimuli and ensures that stimuli relevant to goals or survival gain priority in utilizing higher-level processing resources (Allport, 1989). Guiding attention in service of goals requires voluntary, or endogenous, shifts of attention, whereas involuntary, or exogenous, attention refers to the automatic orienting to highly salient stimuli (Posner, 1980). Initially, researchers suggested that voluntary and involuntary attention were two modes of orienting a unitary attention system (Posner, 1980; Yantis and Jonides, 1990). However, a unitary attention mechanism fails to accommodate the evidence for dissociations between endogenous and exogenous attention. For example, exogenous attention appears to be automatic, stimulus-driven, and resistant to distractors (Jonides, 1981; Prinzmetal et al., 2009; Yantis, 1993), whereas endogenous attention is slower, goal-driven, and more susceptible to distraction (Müller and Rabbitt, 1989; Prinzmetal et al., 2009; Warner

et al., 1990; Wright and Richard, 2000). Endogenous attention builds up over time and can be sustained at the desired location for long periods (Müller and Rabbitt, 1989; Prinzmetal et al., 2009; Warner et al., 1990; Wright and Richard, 2000), whereas exogenous attention is thought to be involuntarily triggered immediately following a salient stimulus, and the mechanism includes the generation of a relative inhibition toward the captured location shortly thereafter (Posner and Cohen, 1984).

Despite the evidence for differences between exogenous and endogenous attention, functional neuroimaging techniques, such as functional magnetic resonance imaging (fMRI) and positron emission tomography (PET), have found strong overlap in the brain regions supporting these types of attention (Corbetta et al., 1993; Corbetta and Shulman, 2002; Kim et al., 1999; Kincade et al., 2005; Nobre et al., 1997; Peelen et al., 2004). Both types of attention activate a bilateral fronto-parietal network, sometimes referred to as the dorsal attention network, even when no saccadic eye movements or spatial orienting are

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required for the attention task (Coull et al., 2000). The dorsal attention network is found to be robustly active for endogenous attentional shifts even in the complete absence of any cue stimuli (Hopfinger et al., 2010). Some have argued that methodological limitations in neuroimaging techniques and differences in stimuli between endogenous and exogenous tasks have made it difficult to examine subtle differences between types of attention using neuroimaging in humans (Pessoa et al., 2003).

Indeed, when other neural techniques have been implemented, differences between endogenous and exogenous attention have been revealed. Work using event-related potentials (ERP) has shown that when endogenous and exogenous attention are in competition with one another, exogenous attention dominates early components whereas endogenous attention dominates later components (Hopfinger and West, 2006). Non-human primate work has demonstrated that lesioning frontal areas (specifically the lateral prefrontal cortex) impairs endogenous but not exogenous attention (Rossi et al., 2007). Conversely, temporarily disrupting the right parietal cortex in humans using repetitive transcranial magnetic stimulation (TMS) reduces exogenous capture but not endogenous attention (Hodsoll et al., 2009). A recent study with transcranial alternating current stimulation (tACS) has also shown a distinction in which stimulation over parietal regions at a frequency within the gamma range facilitates endogenous attention but does not affect exogenous orienting (Hopfinger et al., 2016).

With this evidence, it has been suggested that although endogenous and exogenous attention may rely on the same neural substrates, the former may be relying on frontal-to-parietal processes while the latter relies on parietal-to-frontal processes (Baluch and Itti, 2011; Li et al., 2010). Some of the strongest evidence for this argument stems from single-unit recording in rhesus macaque monkeys. Buschman and Miller (2007) trained macaques on both endogenous and exogenous attention tasks, and then recorded neuronal firing from electrodes in both frontal eye fields and the lateral intraparietal area during attention tasks. The critical results showed that parietal areas peaked before frontal areas in exogenous attention, whereas the reverse was true for endogenous attention.

It is the aim of the current study to expand upon previous neuroimaging work to explore whether differences between endogenous and exogenous attention can be observed in the human dorsal fronto-parietal network, specifically the frontal eye fields (FEF) and intraparietal sulcus (IPS). To do so, we optimized conditions to be more sensitive to revealing the differences between these types of attention. For example, some previous fMRI studies have used arrows to cue endogenous attentional shifts; however, previous work has shown that because these stimuli are over-trained in our daily lives, arrows produce involuntary, exogenous attentional shifts even when the arrow-cue is non-predictive (Ristic et al., 2007). In the present study, fMRI data was collected while participants performed an endogenous attention task (cued by a color change to the fixation cross), a separate exogenous attention task (cued by a peripheral luminance onset), and a separate antipredictive task that was intended to combine elements of both exogenous and endogenous attention. All task goals and target stimuli were matched across conditions to minimize differences between conditions. Analyses focused upon key regions of interest (the FEF and IPS) within the dorsal fronto-parietal attention network that have previously been examined in multi-unit recording and neuroimaging work. Given the a priori regions of interest could be acquired in relatively few slices, we chose to collect only partial brain volumes, centered on these regions. This allowed us to use a faster rate of acquisition (500 msec TR, as opposed to the more typical 2000 msec TR), which potentially reduces the influence of spurious noise and provides a robust dataset without the need for excessively long scanning sessions, which have been linked with decline in task adherence and performance (Lim et al., 2011). This faster scanning rate also permits a more fine-grained analysis of the time course of neural activity, which although still limited by the relatively sluggish hemodynamic response, was analyzed here to further

probe differences in this network across conditions. A goal of the present study was thus to use the more detailed time course to investigate if findings from non-human electrophysiology suggesting a more frontally dominated pattern for endogenous attention and a more parietally-dominated pattern for exogenous attention (Buschman and Miller, 2007) could be revealed with fMRI in healthy human participants.

2. Methods

2.1. Participants

Twenty healthy college students (11 female, mean age 23.8) participated in this study and were paid \$20/hour. Participants were screened to ensure that they were free of neurological or psychiatric disorders, had no metal in their bodies, were not currently pregnant, and had normal or corrected to normal vision. All procedures were approved by the Institutional Review Board at the University of North Carolina at Chapel Hill, and all participants provided written informed consent.

2.2. Task

Participants viewed images on a translucent screen through an angled mirror attached to the head coil. Commercial software (Neurobehavioral Systems, San Francisco, CA) was used to present stimuli and record responses and reaction times. Before completing experimental runs, all participants completed a behavioral training session to ensure high performance accuracy. There were three different cue conditions: an endogenous cue condition (Endo), an exogenous cue condition (Exo), and an antipredictive exogenous cue condition (Anti). Conditions were blocked such that participants only completed one type of cue condition per run. Order of conditions was counterbalanced across participants. Each run consisted of 64 trials. Participants completed two runs of each type of cue condition for a total of 6 functional runs and 384 trials per participant.

2.2.1. Endo condition

This condition was designed to engage voluntary attention using a color-change cue of the fixation cross. The sequence of events on a typical endogenous trial is illustrated in Fig. 1 (left column). Each trial began with a 600–700 ms presentation of a preview display that consisted of a black fixation cross and two outline black boxes (3.2° high and 3.2° wide) against a dark gray background. The two black boxes appeared above middle line, and their inner edge was either 5° left or right of the central fixation cross. Participants were asked to keep their eyes fixated on the central cross. After a 600–700 ms delay, the central cross flashed green or red for 100 ms. Half of participants were instructed that the target was more likely to appear on the left for green and on the right for red and the other half were told the reverse. There was an inter-stimulus interval (ISI) of 600–900 ms between the cue and target. Following the ISI, the target, a black-white checkerboard, appeared in one of two peripheral boxes for 150 ms. The center of the checkerboard target was a solid gray square for 50% of trials, and alternated between white and black for 50% of trials. Participants were asked to report whether there was a gray center or not by pressing a keyboard as quickly and as accurately as possible. The inter-trial interval (ITI) varied between 4 and 11 s. Within a run, the target was absent for 16 of the 64 trials - these “catch trials” ensured that the subject would wait for the target to appear before responding. Of the 48 trials in which the target was displayed, 40 trials (83%) were “valid” trials, in which the target appeared in the expected location (i.e. the location indicated by the color of the cue), and 8 trials (17%) were “invalid” trials in which the target appeared in the unexpected location (i.e. opposite the location indicated by the cue).

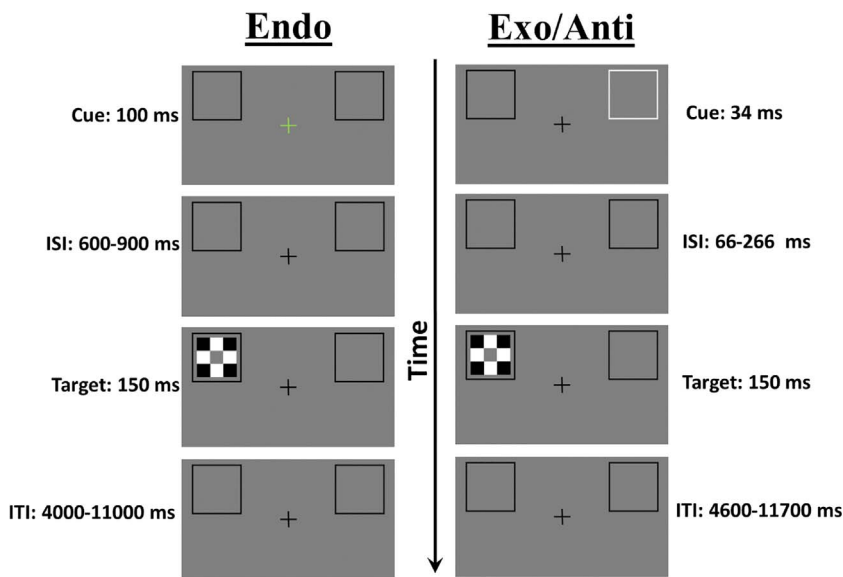


Fig. 1. Trial sequence and stimuli. Left column: Endogenous cue condition. The cue was a color change of fixation cross from black to either red or green. Twenty-five percent of trials were “catch” trials that did not contain any target stimulus. On the 75% of trials with targets, 83% of those were valid trials and 17% were invalid trials. Right column: Exogenous and Anti-predictive conditions. As in the endogenous condition, 25% of trials were “catch” trials that contained no target stimulus. For the 75% with targets, 50% of those were valid trials and 50% were invalid trials in the Exogenous Condition. In the Antipredictive condition, the target occurred at the cued location on 17% of trials, and at the uncued-location on 83% of trials.

2.2.2. Exo condition

This condition was designed to engage involuntary attention using a peripheral flash that was salient, but non-predictive of target location. The sequence of events in a typical exogenous trial is illustrated in Fig. 1, right column. Just as with the Endo condition, each trial began with 600–700 ms presentation of a preview display with a black fixation cross and two outline black boxes against a dark gray background. After the preview display, one of two black boxes turned white for 34 ms, which was the cue. There was an ISI of 66–266 ms before the target appeared for 150 ms. The checkerboard target and task goals were the same as for the Endo condition. The ITI varied between 4600 and 11,700 ms. The ITI was different in the Exo and Endo conditions so that the overall trial length would be identical in each of the conditions. This ensured that the total length of each run, and the total number of trials within each run, were identical across the conditions. As with the Endo condition, catch trials, in which no target appeared, accounted for 16 of the 64 trials in a run. For the 48 trials in which targets appeared, 50% of trials were “valid” trials in which the cue and target appeared in the same location (i.e. the checkerboard appeared in the box that flashed white). The other 50% of target present trials were “invalid” trials, in which the target appeared in the opposite location from the cue. Consistent with the presentation, participants were told that the white box was completely uninformative with regards to the location of the target, and therefore should be ignored.

2.2.3. Anti condition

This condition was designed to engage both types of attention by using a salient exogenous cue to signal participants to voluntarily orient their attention to the uncued-location. The Anti condition shared all of the same physical stimuli, timing, and target response as the Exo condition (shown in Fig. 1, right column). However, in the Anti condition, participants were instructed to use the cue, the white box, to voluntarily shift attention to the opposite box. Again, within a run, 16 of the 64 trials were catch trials in which the target was absent. Consistent with instruction, in the 48 trials with a target, 40 trials (83%) were “valid” trials in which the target appeared in the *opposite* location of the cue and 8 trials (17%) were “invalid” trials in which the target appeared in the same location as the cue.

To address the primary question of interest, which regarded comparisons between endogenous and exogenous attention, we elected to use data from the Anti condition to determine the locations for ROIs but not include the Anti data itself in any tests of statistical inference for this paper. Because the Anti condition was designed to engage both

endogenous and exogenous attention, this method of defining ROIs allowed us to not bias our selection of regions by using a task that only shared processing with one of our conditions. Furthermore, by using the Anti condition to define the ROIs, we avoid the problem of using the same set of data to define ROI locations and to test hypotheses, which previous work has shown can yield invalid statistical inferences (Kriegeskorte et al., 2009).

2.3. MRI scanning protocol

Images were collected using a 3T Siemens Trio Total imaging matrix (Tim) MRI system at the University of North Carolina Biomedical Research Imaging Center. Partial brain imaging focused on the dorsal fronto-parietal network including the frontal and parietal areas implicated in previous primate work. Functional images included 10 transverse slices ($4.0 \times 4.0 \times 6.0 \text{ mm}^3$ resolution), collected interleaved inferior to superior, with the most superior images just beneath the top of the skull (See Fig. S1 in Supplementary Materials). Images were acquired using a T2-weighted echo-planar imaging (EPI) sequence (TR = 500 ms, TE = 27 ms, flip angle = 90 degrees), and the first two scans were discarded to allow for magnetic field stabilization. Participants completed 6 fMRI runs (7.3 min each) with 2 runs for each of the three attention conditions. A structural scan was acquired for each participant (T1-weighted; TR = 1900 ms; TE = 2.32 ms; flip angle = 9, FOV = 230 mm, 192 slices, $.9 \times .9 \times .9 \text{ mm}^3$ resolution).

2.4. fMRI analyses

fMRI data were preprocessed using SPM12 (Wellcome Department of Imaging Neuroscience, University College London, UK). Preprocessing included slice-time correction and realignment using a two pass procedure in which images were realigned to register to the mean of the images created after the first realignment. The mean image constructed from realignment was used to determine parameters for spatial normalization into the standard EPI-template and smoothed with an 8 mm (FWHM) isotropic kernel.

For the analysis of task-dependent activation, each of the six fMRI runs consisted of covariates of interest classifying each trial in terms of *Cue Location* (left vs. right) and *Trial Type* (catch, invalid, valid). Each of the conditions (Endo, Exo, and Anti) were represented in two of the six runs. To account for effects of no interest in the model, six movement parameters obtained during realignment were included as covariates of no interest for each run, as well as one covariate for each run

incorporating an overall intercept to the model. Each regressor was convolved with canonical response hemodynamic function and entered into the general linear model (Friston et al., 1995). Contrast maps for each subject were entered into a second-level analysis.

2.5. Region of interest analyses

Region of interest (ROI) analyses were conducted on *a priori* selected regions (bilateral FEF, and bilateral IPS) of the dorsal attention network, to examine possible differences between the Endo and Exo conditions. Local maxima obtained from the Anti condition was used to define the location of the center of 8 mm radius ROI spheres (Table 2), because this condition was separate from the conditions to be analyzed, yet shared some features with both conditions (the same stimuli and timing as the Exo condition; the same predictive validity as the Endo condition). This allowed us to investigate regions that were potentially active in both tasks, to see if even when the same brain regions are active, they may show a different balance of activity when involved in different types of attentional orienting. ROIs were generated using MarsBar (Brett et al., 2002).

The ROI analyses were conducted using contrast values of each event-type against baseline with activity averaged across all voxels of the ROI for each participant. Each participant's averaged contrast values for each event-type were then exported to SPSS 23.00 and entered into a $2 \times 2 \times 2 \times 2$ repeated measures analysis of variance (ANOVA) with the factors hemisphere (left, right), condition (Endo, Exo), ROI (FEF, IPS), and validity (Valid, Invalid). Any significant results were probed using least-significant difference (LSD) pairwise comparisons. In addition to extracting mean activation, time series data were extracted from each ROI using a finite impulse response (FIR) model in MarsBar.

In addition to the standard trials analyzed above, which included cue and target stimuli, we also performed follow-up analyses on the rare catch trials. These trials potentially allow for analysis of attentional control processing without any target processing; however, it should be noted that they do not represent an entirely pure measure of attention orienting. Specifically, previous research provides reason to treat results from these rare catch trials with caution, because they are typically of lower power (due to smaller number of trials) and because these trials are associated with additional processing, not associated with the standard trials (Busse and Woldorff, 2003). This additional processing is thought to be related to these catch trials violating the typical pattern of the majority of trials; therefore neural responses to these “oddball” trials can reflect some degree of expectancy violation, subject realization that the trial type is different, and the cancellation of motoric response preparation. To balance statistical power with isolation of attentional control, the standard trials (with higher power) were analyzed first, with catch trials submitted to subsequent confirmatory follow-up analyses.

2.6. Time series extraction

While average ROI activity differences between endogenous and exogenous attention was of primary interest for this study, looking at BOLD responses for these conditions over time can help to better characterize differences. For example, Buschman and Miller (2007) demonstrated in non-human primates that parietal areas peak in activity before frontal areas for exogenous attentional shifts but the reverse timing is true for endogenous attentional shifts. To better describe temporal characteristics of each condition in the present study, an FIR model was used to extract time course data from each ROI, with no assumptions about the shape of activations. Corresponding with the repetition time (TR), 500 ms time bins were used and examined across 32 time points (16 s) following the presentation of the cue. To test differences across time series, activity around the peak of the hemodynamic response (from 3 s to 8 s) were entered into a $2 \times 2 \times 2 \times 2 \times 11$ repeated measures analysis of variance (ANOVA)

with the factors hemisphere (left, right), condition (Endo, Exo), ROI (FEF, IPS), validity (Valid, Invalid), and time point (every 500 msec, from 3 through 8 s post-cue presentation). Follow-up analyses were also performed on the rare catch trials.

2.7. Behavioral analyses

Although behavioral responses were not the primary interest of this study, manual responses to the targets were collected and these data were inspected to confirm engagement with the task. Because task instructions mentioned both speed and accuracy, participants may have differed in the degree to which they focused on speed versus accuracy in their overt responses. Therefore, we used a metric, inverse efficiency, which is used to account for both accuracy and reaction time in analyzing behavioral responses. Inverse efficiency is calculated for each subject as the subject's average response time on correct trials of a given trial type divided by the subject's accuracy for that trial type as a fraction out of 100. These scores were then entered into a 3×2 repeated measures ANOVA with the factors of condition (Endo, Exo, Anti) and validity (Valid, Invalid).

3. Results

3.1. Behavioral results

To account for potential speed-accuracy trade-offs, inverse efficiency was examined. Inverse efficiency weights reaction time with accuracy such that better performance is indicated by lower numbers. Analyses of the inverse efficiency revealed a main effect of condition, $F(2, 38) = 44.306$, $p < .001$, such that participants showed overall higher inverse efficiency (i.e. worse overall performance) in the Endo condition compared to both the Anti and Exo conditions, p 's $< .001$ (Table 1; Supplemental Table S1 presents separate reaction time and accuracy scores). There was a trend toward overall worse performance in the Anti condition than in the Exo condition, $p = .069$. There was also a main effect of validity, $F(1,19) = 174.402$, $p < .001$, such that participants performed better on the valid than the invalid trials, mimicking typical validity effects in attention experiments. These were qualified by a significant condition by validity interaction, $F(2, 38) = 143.777$, $p < .001$. Performance was better for valid than invalid trials, for both Endo, $p < .001$, and Anti conditions, $p < .001$, providing evidence that in both of these attention conditions, participants were able to voluntarily orient to the expected location in preparation for the target. In the Exo condition, a negative validity effect was revealed, with performance being better for invalid than for valid trials, $p = .032$. Such a negative validity effect corresponds with inhibition of return (IOR), an automatic consequence of involuntary attention (Posner and Cohen, 1984). Although some models have assumed that a facilitation at the cued-location should precede IOR, a number of studies have found that IOR is a more robust marker of exogenous attention, as the initial facilitation is not always found (Chica et al., 2014; Klein, 2000; Tassinari et al., 1994). As Klein (2000) explained, some data are consistent with an inhibition mechanisms being initiated with the initial onset of the peripheral cue, with other factors determining if this effect is overridden in overt response measures by an additional facilitative process at the cued location. The present finding of a

Table 1
Inverse efficiency (IE) measures across conditions. Mean(SD).

	Valid	Invalid	Validity Effect
Exo	574.819 (19.469)	552.824 (53.639)	−21.997 [*]
Endo	542.158 (18.123)	785.388 (49.541)	243.231 [*]
Anti	539.0245 (48.732)	631.663 (76.597)	92.639 [*]

^{*} Indicates significant at $p < .05$.

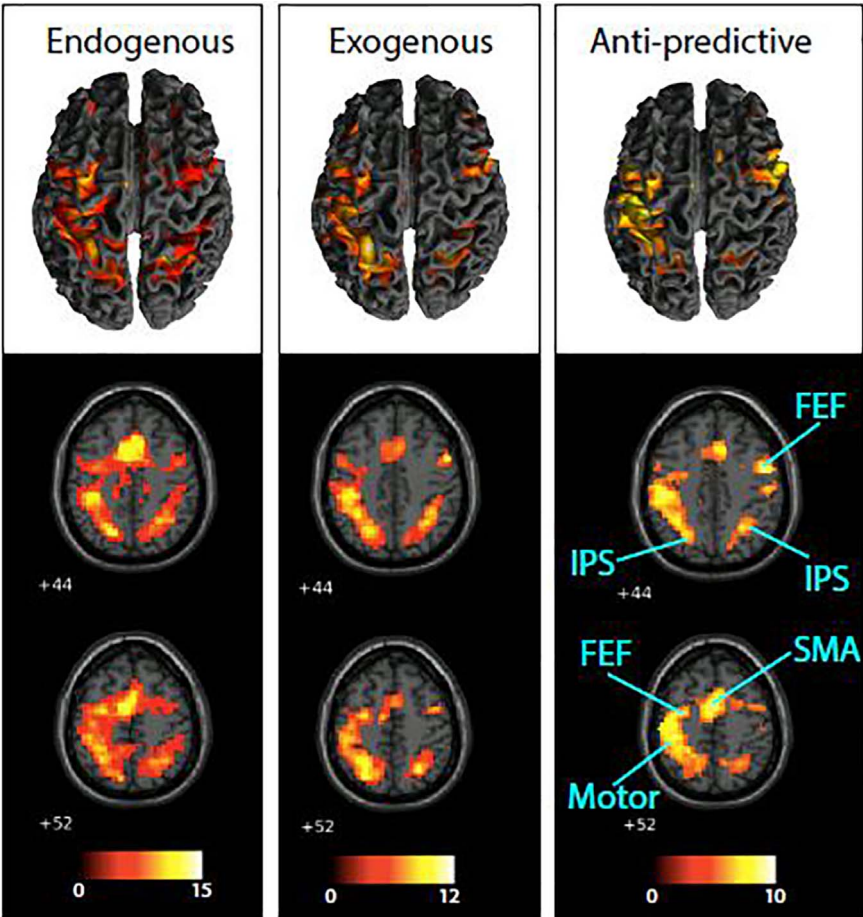


Fig. 2. Activity from the 3 attention conditions. All maps have threshold of $p < .05$ FWE-corrected.

Table 2
ROI locations in MNI space and activity (presented as t-score values) across conditions.

ROI	Coordinates			Anti	Endo	Exo
	x	y	z			
Left FEF	−39	−7	56	10.02	9.02	7.03
Right FEF	45	−1	41	11.40	9.73	6.47
Left IPS	−24	−64	47	9.05	12.96	10.54
Right IPS	27	−55	44	9.00	10.72	8.24

FEF, frontal eye field; IPS, intra-parietal sulcus. Coordinates: x, left/right; y, anterior/posterior; z, inferior/superior in the reference frame of the single subject canonical brain in SPM12. All activations significant at $p < .05$ FWE-corrected.

significant behavioral effect following the exogenous cue suggests that this cue was indeed strong enough to trigger involuntary orienting. In addition, the robust validity effect in the Anti condition provides further evidence that this cue did not hold attention long, as participants were able to rapidly reorient to the expected target location when the target location was predicted by the cue.

3.2. fMRI full volume results

For each condition (Endo, Exo, and Anti), a planned t-contrast was used to compare activation to an implicit baseline for all standard trials. Each condition showed activation of the fronto-parietal network, including FEF and IPS, as well as SMA and motor areas, as shown in Fig. 2. Six directed t-contrasts were conducted to examine whole brain activation differences between conditions (Endo > Exo, Exo > Endo, Endo > Anti, Anti > Endo, Exo > Anti, Anti > Endo). In these

comparisons, no voxels exceeded the threshold of $p < .05$ at FWE whole-brain corrected, indicating no overall activation differences between conditions. This was not surprising given previous research indicating a large degree of overlap in activation of the fronto-parietal networks across voluntary and involuntary attention conditions (Corbetta et al., 1993; Corbetta and Shulman, 2002; Kim et al., 1999; Kincade et al., 2005; Nobre et al., 1997; Peelen et al., 2004).

3.3. fMRI region of interest (ROI) results (overall)

Of primary interest was whether neural activity for these conditions differed within the fronto-parietal network (bilateral FEF and IPS). Therefore a more sensitive set of ROI analyses were performed comparing ROI activity between the Exo and Endo conditions using the ROI locations defined from the ANTI condition (to avoid testing the same data that was used to define the ROIs). Coordinates and T-scores for each region of interest are shown in Table 2. Results of the repeated measures ANOVA revealed no main effect of condition, $F(1, 19) = .794$, $p = .384$, and no overall difference between the FEF and IPS regions, $F(1, 19) = .763$, $p = .393$. There was a main effect of validity, such that invalid trials showed greater task-related activity in the selected ROI's than valid trials $F(1, 19) = 37.844$, $p < .001$. Importantly, there was a main effect of brain hemisphere, in that task-related changes in activity were greater in ROI's located in the left hemisphere than the right hemisphere, $F(1, 19) = 23.752$, $p < .001$. This finding is in agreement with recent studies showing greater effects of task-based connectivity changes in the left versus right hemisphere attention networks (Meehan et al., 2017). There was a significant two-way validity by hemisphere interaction, $F(1, 19) = 7.821$, $p = .012$, which showed that while the validity effect was in the same direction for both hemispheres

(invalid > valid), the effect was larger in the left hemisphere (mean difference = 2.071) than in the right hemisphere (mean difference = 1.647). Most critical and relevant for the specific aim of our investigation, there was a significant three-way *condition by region by hemisphere* interaction, $F(1,19) = 6.251$, $p = .022$. No other two-way, three-way, or four-way interactions were significant (all p 's > .02). To further characterize this critical significant three-way interaction, the data was further submitted to two separate $2 \times 2 \times 2$ repeated measures ANOVAs with factors condition (Endo, Exo), validity (Valid, Invalid), and region (FEF, IPS), separately for the left and right hemispheres, as described below.

3.3.1. Left hemisphere ROIs

Results of analyses in the left hemisphere again showed similar main effects as the overall ANOVA, such that there was no effect of condition $F(1, 19) = .859$, $p = .366$, and no effect of region $F(1, 19) = .165$, $p = .690$, with a significant effect of validity $F(1, 19) = 35.886$, $p < .001$. Critically, within the left hemisphere, there was a significant region by condition interaction, $F(1, 19) = 7.243$, $p = .014$. As shown in Fig. 3 (left panel), this interaction was driven by the fact that in the Endo condition FEF was significantly more active than IPS, $p = .030$; in the Exo condition, the pattern was in the opposite direction (IPS more active than FEF), but was not significant, $p = .120$. There were no other significant two-way or three-way interactions, (all p 's > .06).

3.3.2. Right hemisphere ROIs

In the Right hemisphere, the main effect of validity, $F(1, 19) = 36.364$, $p < .001$, and the lack of an overall main effect of region, $F(1,19) = 1.271$, $p = .274$, matched the patterns shown in the left hemisphere and in the overall ANOVA. Critically, however, and unlike the left hemisphere, there was no condition by region interaction, $F(1, 19) = .007$, $p = .936$ in the right hemisphere, as the balance of activity between FEF and IPS was the same for Endo and Exo (Fig. 3, right panel). There were no other significant main effects or interactions (all p 's > .1) in the right hemisphere.

3.3.3. Catch trial ROI analysis

Although there were only a relatively small number of catch trials, and therefore the data may be less robust and powerful, we conducted a follow-up analysis on these trials to examine whether the results found above may have been due to target processing. A repeated measures ANOVA was conducted on catch trials alone, using the factors of hemisphere (left, right), condition (Endo, Exo), and ROI (FEF, IPS). This catch trial analysis replicated results found in the analysis above,

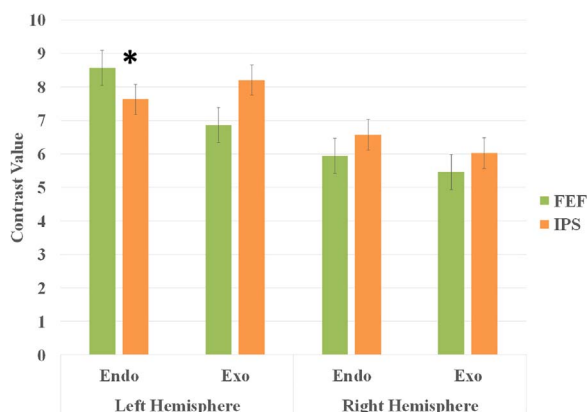


Fig. 3. ROI activity as a function of condition and hemisphere for the Frontal Eye Fields (FEF) and Intraparietal Sulcus (IPS) for target present (valid and invalid) trials. * indicates where the difference between FEF and IPS is significant, $p < .05$. Although there was no 3-way interaction with validity, Supplemental Fig. S2 shows these same data broken down further by validity to illustrate that the pattern was similar for both valid and invalid trials.

revealing a main effect of hemisphere, $F(1,18) = 8.050$, $p = .011$, such that ROI activity in the left hemisphere was overall greater than activity in the right hemisphere. There was also no main effect of condition ($p = .965$) or region ($p = .844$). The primary interaction of interest, the 3-way *condition by hemisphere by region* interaction, was again significant in the analysis of catch trials alone, $F(1, 18) = 9.010$, $p = .008$. No two-way interactions reached significance (all p 's > .1). Further analysis of the three-way interaction replicated that there was a significant condition by region interaction in the left hemisphere, $F(1,18) = 4.401$, $p = .050$, but not in the right hemisphere, $p = .066$. No main effects of condition or region reached significance when tested separately in either hemisphere (p 's > .1). The condition by region interaction in the left hemisphere indicated that the FEF was overall more active than the IPS in the Endo condition, $p = .007$, with no difference in the Exo condition, $p = .690$ (Fig. 4 left panel). No differences between conditions within region, or between regions within condition, reached significance in the right hemisphere (p 's > .1; Fig. 4 right panel).

3.4. fMRI time series analyses

The time series were examined to explore whether the condition by region interactions shown in the left hemisphere in overall activity were due simply to differences in peak activity, or rather may be due to a more complex function over time. The analysis focused on time points just before and after the peak, with 500 ms intervals from 3 to 8 s post-cue presentation. The primary interaction of interest was the 4-way condition by region by hemisphere by time point interaction, which was revealed to be significant, $F(4,954, 94.131) = 3.944$, $p = .003$. Furthermore, this 4-way interaction was not qualified by validity ($p = .280$ for the 5-way interaction). Because any main effects or 2-way or 3-way interactions including hemisphere, region, condition, and time point are qualified by this 4-way interaction, they are not interpreted at this level (statistics for all main effects and interactions are provided in Table S2 in the Supplementary Materials). This interaction was further probed by examining the left and right hemisphere activity separately, as discussed in the follow-up ANOVA's described below.

3.4.1. Left hemisphere time series results

To investigate the 4-way interaction, a follow-up ANOVA with factors of condition, validity, region, and time point was conducted in each hemisphere separately. Critically, within the left hemisphere, a significant condition by region by time point interaction was revealed, $F(3,352, 63.695) = 4.916$, $p = .003$. In order to better interpret this interaction, follow-up ANOVAs were conducted both within each of the Endo and Exo conditions and within the regions of FEF and IPS. Because the ANOVA controls for type-I error rate, any effects that reached

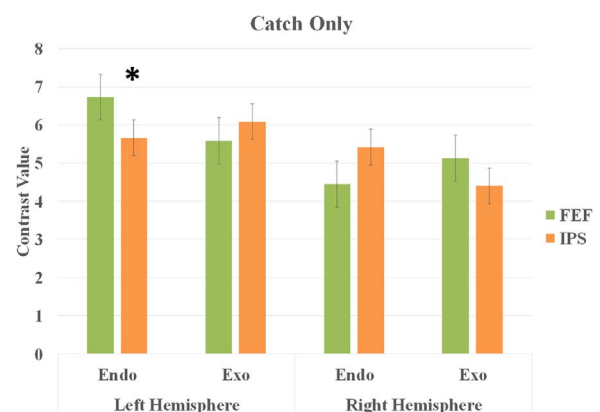


Fig. 4. ROI activity as a function of condition and hemisphere for the Frontal Eye Fields (FEF) and Intraparietal Sulcus (IPS) for catch trials. * indicates where the difference between FEF and IPS is significant, $p < .05$.

Table 3
Results of time-series pairwise comparisons in target present trials in left hemisphere regions.

	Time in seconds	Mean difference	t-value	p
Endo (FEF > IPS)				
	3	.052	5.439	< .001
	3.5	.071	5.942	< .001
	4	.075	7.202	< .001
	4.5	.075	5.404	< .001
	5	.072	5.391	< .001
	5.5	.072	4.027	.001
	6	.052	3.431	.003
	6.5	.045	2.887	.009
	7	.027	1.765	.094
	7.5	.004	.277	.785
	8	.017	1.177	.254
Exo (IPS > FEF)				
	3	-.024	2.040	.056
	3.5	-.024	1.611	.124
	4	-.026	1.251	.226
	4.5	-.012	.471	.643
	5	-.001	.045	.965
	5.5	.024	.889	.385
	6	.040	1.512	.147
	6.5	.056	1.972	.063
	7	.057	2.443	.025
	7.5	.050	2.489	.022
	8	.037	2.310	.032
FEF (Endo > Exo)				
	3	-.035	1.624	.121
	3.5	-.025	.893	.383
	4	-.016	.545	.592
	4.5	.008	.252	.804
	5	.038	1.268	.220
	5.5	.064	2.002	.060
	6	.095	3.249	.004
	6.5	.101	3.358	.003
	7	.095	3.818	.001
	7.5	.064	3.114	.006
	8	.048	2.208	.040
IPS (Exo > Endo)				
	3	.063	4.294	< .001
	3.5	.072	3.841	.001
	4	.065	3.078	.006
	4.5	.056	2.525	.021
	5	.035	1.655	.114
	5.5	.032	1.522	.144
	6	-.003	.169	.867
	6.5	< .001	.002	.998
	7	-.010	.550	.589
	7.5	-.010	.569	.576
	8	.005	.281	.782

significance when conducting follow-up ANOVAs were probed using least-significant difference (LSD) pair-wise comparisons. Within the Endo condition, a significant region by time point interaction was found, $F(2.806, 53.313) = 8.170$, $p < .001$. This interaction was further probed with paired samples tests, showing that in the Endo condition, activity in the FEF was greater than the IPS early in time, from 3 s to 6.5 s post cue presentation (p 's < .01; Table 3, Fig. 5, top). In the Exo condition, a significant region by time interaction, was found $F(2.134, 40.555) = 11.445$, $p < .001$; however, this was driven instead by activity in the IPS being greater than activity in the FEF, and at a later time, from 6.5 s to 8 s post cue presentation (p 's < .05; Table 3). Alternatively, examining differences within each region revealed a significant condition by time point interaction in both the FEF, $F(3.058, 58.106) = 21.198$, $p < .001$, and the IPS, $F(3.388, 64.366) = 15.757$, $p < .001$. Specifically, in the FEF, activity in the Endo condition was greater than activity in the Exo condition late, from 6 through 8 s post cue onset (p 's < .05), whereas in the IPS, activity in the Exo condition was greater than activity in the Endo condition early, from 3 through 4.5 s post cue presentation (p 's < .05; Table 3).

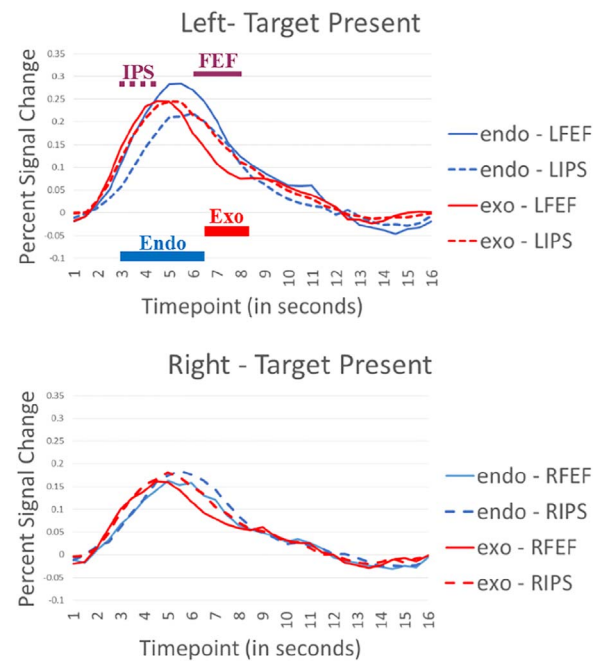


Fig. 5. Peri-stimulus Time Histogram (PSTH) of target present trials shown as percent signal change in TR's, with each TR = 500 ms, in the left hemisphere (top) and the right hemisphere (bottom). Blue bar indicates time points at which FEF and IPS regions differ significantly in activity in the Endo condition; Red bar indicates time points in which FEF and IPS regions differ significantly in the Exo condition; Dotted purple line indicate time points at which Exo and Endo differ significantly in activity in the Endo condition; Solid purple line indicate time points at which Exo and Endo differ significantly within the FEF. Significance assessed with Fisher's protected least significant difference tests. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.4.2. Right hemisphere time series results

Mimicking results from the analysis of average activity, within the right hemisphere, there was no main effect of region or condition, no condition by region interaction, and no condition by region by time point interaction (p 's > .1; Fig. 5, bottom). The analysis did show a condition by time point interaction, $F(3.8880, 73.724) = 13.367$, $p < .001$: follow-up tests using LSD showed that at one time point, 7.5 s post stimulus presentation, the Endo condition yielded more activity than the Exo condition, averaging across the regions of FEF and IPS ($p = .043$). Additionally, a region by time point interaction, $F(3.593, 68.261) = 4.74$, $p = .004$, showed that at one time point, 7 s post cue presentation, there was overall more activity in the IPS than the FEF when averaging across Endo and Exo conditions, $p = .021$. Because the condition by region by time point interaction did not reach significance, no follow-up pairwise comparisons (of conditions within region or regions within condition) were conducted within the right hemisphere regions.

3.4.3. Catch trial time series analysis

Similar to the analysis of average ROI activity above, catch trials were analyzed separately to test whether results were due to attentional control specifically without the presence of a target. Before describing those results, though, we note that visual inspection of the time course plots reveals that the shape of the activity plot over time for these catch trials is noticeably different from the standard event-related hemodynamic response (Fig. 6). Specifically, as opposed to a typical rather sharp peak of activity at ~6 s, the catch trials show a much more sustained response. This may be due to participants treating these trials differently, as they are forced to anticipate the target for a longer period of time, and ultimately realize it is a different type of trial, with no target appearing. As described in the methods section, such differences in catch trials have previously been documented (Busse and Woldorff,

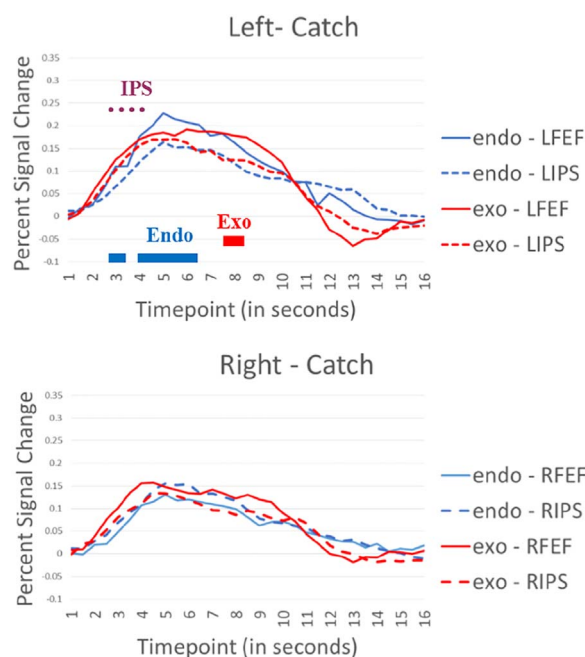


Fig. 6. Peri-stimulus Time Histogram (PSTH) of catch trials shown as percent signal change in TR's, with each TR = 500 ms, in the left hemisphere (top) and the right hemisphere (bottom). Blue bar indicates time points at which FEF and IPS regions differ in activity in the Endo condition; Red bar indicates time points in which FEF and IPS regions differ in the Exo condition; Dotted purple line indicate time points at which Exo and Endo differ in activity in the Endo condition. Of these, only the region differences in the Endo condition from 4 to 5.5 s survive a Bonferroni correction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2003), and analyses on these trials are of lower power than the target present trials. Therefore, we present the catch trial data as additional information, with the caveat that activity within these trials, especially during the later stages of the activity time course, may include additional top-down processes not associated with the initial attention-orienting processing of interest here. Within the catch trials alone, an ANOVA with the factors of condition, hemisphere, region, and time point was conducted. A main effect of time point, $F(2.354, 44.733) = 11.463$, $p < .001$, indicated that the most activity was from 5 s to 7 s post stimulus onset, with no differences in activity level between time points in this 2 s window (p 's $> .05$). The main effect of hemisphere, $F(1, 19) = 15.390$, $p = .001$, and the condition by hemisphere by region interaction, $F(1, 19) = 6.748$, $p = .018$, were the same as was found in the average activity analysis, and no effects interacted with time, (p 's $> .05$).

As reported above, both visual inspection of the catch trial time course and statistical tests of the main effect of time point indicate a more platykurtic distribution over time rather than a typical peak, which may be due to different processing or low power on these trials. For this reason, we chose to conduct a more restricted set of paired-sample t -tests, across only the specific time ranges that were significantly different in the target-present analysis. To further account for multiple tests being conducted without the protection of a significant interaction, the more stringent Bonferroni correction was used here rather than least-significant difference. The results are reported both with and without Bonferroni correction to provide a more complete view of the data. The Bonferroni corrections here differ across regions/conditions, because within each region/condition, we only tested the time points that were significant in the original timeseries analyses, as described below. Testing across 3–6.5 s post-cue presentation in the Endo condition, FEF is significantly more active than IPS at 3 s and at all time points from 4 to 6.5 s (inclusive) post-cue presentation (p 's $< .05$; Table 4). After Bonferroni correction (8 time points; Bonferroni

Table 4

Results of time-series paired t -test in catch-trials in left hemisphere regions.

	Time in seconds	Mean difference	t-value	p
Endo (FEF > IPS)	3	.044	2.627	.017
	3.5	.051	.602	.554
	4	.056	3.434	.003
	4.5	.053	3.728	.001
	5	.065	3.586	.002
	5.5	.082	3.410	.003
	6	.055	2.520	.021
	6.5	.056	2.484	.022
Exo (IPS > FEF)	6.5	.045	1.601	.126
	7	.042	1.470	.158
	7.5	.060	2.490	.022
	8	.053	2.395	.027
FEF (Endo > Exo)	6	.016	.559	.582
	6.5	.015	.505	.620
	7	-.010	.312	.759
	7.5	-.002	.070	.945
	8	-.015	.548	.590
IPS (Exo > Endo)	3	.035	2.104	.049
	3.5	.043	2.190	.041
	4	.037	1.970	.064
	4.5	.025	1.430	.169

corrected threshold $p < .0063$), FEF is still significantly more active than IPS at 4 through 5.5 s (p 's $< .005$). Testing across 6.5–8 s in the Exo condition, the FEF is significantly more active than the IPS at 7.5–8 s (p 's $< .05$; Table 4); however, these differences did not reach significance after Bonferroni correction (p 's $> .02$; Bonferroni corrected threshold $p < .0125$ for 4 time points). When comparing conditions within the FEF from 6 through 8 s, there were no differences between the Endo and Exo conditions with or without Bonferroni correction (p 's $> .5$; Table 4). When comparing activity within the IPS from 3 through 4.5 s, Exo shows significantly more activity than Endo from 3 to 3.5 s (p 's $< .05$; Table 4); however, these differences were no longer significant after Bonferroni correction (p 's $> .04$; Bonferroni corrected threshold $p < .0125$ for 4 time points).

4. Discussion

This study aimed to test the hypothesis that the pattern of neural activity within the dorsal fronto-parietal network is different for top-down voluntary attention versus bottom-up involuntary attention. Our results provide new evidence that differences between endogenous and exogenous attention can be observed even within a neural system, the human dorsal attention network that is engaged by both types of attention. Specifically, within the left hemisphere, frontal activity was greater than parietal activity in the endogenous condition whereas the exogenous condition showed the opposite pattern. This supports previous work from non-human animal studies, as well as human lesion and ERP studies, suggesting that endogenous attention is more frontally-dominated whereas exogenous attention is more parietally-dominated (Baluch and Itti, 2011; Buschman and Miller, 2007; Li et al., 2010), and critically extends this to human neuroimaging findings.

A robust result was the hemispheric differences found here - specifically, that the differential balance of power between frontal and parietal areas for endogenous versus exogenous attention was driven by effects in the left hemisphere. This finding extends previous neural evidence from non-human primate data that recorded from only a single hemisphere, or did not report separate results for the two hemispheres. A critical implication of the current study is that taking hemisphere into consideration is essential when evaluating the potential differences in neural mechanisms between voluntary and

involuntary attention in humans. In addition to the differences in the balance of frontal versus parietal activity, left-hemisphere activity was generally larger and showed an exaggerated effect of reorienting (more activity for invalid than valid trials) compared to the right hemisphere, for both voluntary and involuntary attention conditions. This is consistent with previous neuroimaging results, but is in opposition to the right-hemispheric dominance found across neuropsychological patient studies (Heilman and Abell, 1980; Posner and Petersen, 1990). Recent evidence suggests that the findings of left hemisphere dominance in neuroimaging studies of attention could be due to the right hemisphere being more tonically active even during “baseline” periods of the fMRI run, whereas left hemisphere areas may be engaged more specifically during the greater attentional demands that occur during the performance of the attention task (Meehan et al., 2017).

Further investigation of the current attention differences using time series analysis show partial replication of the results from Buschman and Miller (2007). Specifically, their finding that activity in frontal areas preceded activity in the parietal areas during endogenous attention parallels our findings in the left hemisphere that endogenous attention activity in the FEF is significantly greater than in the IPS in an early time window, from 3 to 6.5 s following the cue. Our finding that activity in the IPS is greater than in the FEF for the exogenous condition is also similar to Buschman and Miller's (2007) finding of a dominant role for parietal activity in exogenous attention; however, we do not see evidence that initiation of the parietal activity precedes frontal activity. Instead, our results suggest that exogenous attention activity in the left frontal and parietal areas do not differ prior to the peak of the hemodynamic response, but rather that parietal activity is more sustained than frontal activity in the exogenous attention condition, with significant differences from 6.5–8 s post-cue presentation. These conclusions are further supported by comparisons of attention conditions within regions. Specifically, frontal activity in the exogenous condition is not sustained as long as in the endogenous condition, as shown by the difference in FEF between conditions emerging at 6–8 s post-cue presentation. Notably, this is around the same time period that the parietal versus frontal difference emerge within the exogenous condition, suggesting this falling edge of the hemodynamic response may contain important information for understanding the different mechanisms of endogenous and exogenous attention. In contrast to the left FEF, differences between endogenous and exogenous attention in the left IPS are seen only at early stages, between 3 through 4.5 s post-cue presentation with no differences after the peak; this supports the suggestion that parietal areas are engaged earlier in exogenous attention than endogenous attention. In summary, these results show that early in time, endogenous attention is frontally-dominated and shows less parietal activity than in exogenous attention, possibly due to a frontal-to-parietal attention mechanism in endogenous attention. However, later in time, exogenous attention is more parietally-dominated, likely due to a less sustained frontal response, as no difference in frontal activity between endogenous and exogenous conditions is observed until after the peak of the hemodynamic response.

There are some limitations of the current study to be considered as well. Because this experiment was designed to focus specifically on the effects of endogenous and exogenous attention within the dorsal attention network, our data collection and analysis did not go beyond these brain regions. As such, we were unable to determine what role regions outside of this fronto-parietal network may be playing, and in particular the role of the ventral attention network. Given that the dorsal and ventral networks have been shown to interact (see Vossel et al., 2014 for review), there may indeed be influences of the ventral attention network on the activity and balance of power observed here. As with any neuroimaging study, we cannot make firm conclusions on the temporal order of these activities or the presence of a direct causal relation between these regions. Despite these limitations, however, we feel that this study makes a novel contribution to the field by better characterizing the differences between endogenous and exogenous

attention even within a network that often appears to be equivalently active when analyzed in the standard way. Understanding the unique versus the shared mechanisms of these different attention processes is an important step in addressing their differential impact in attentional dysfunction, ranging from transient dangers (e.g., distracted driving) to sustained impairments (e.g., attention deficits or harmful attention biases).

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.neuropsychologia.2018.02.006>.

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